

Determinants of Insurance Premiums: Lifestyle Choices vs. Demographic Factors

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1. Introduction

Within 2023, healthcare spending in the United States increased by 7.5%, totaling to 4.9 trillion USD; this healthcare spending increase parallels increases in the cost of health insurance. Recent rates of change in health insurance premium cost has exceeded that of wages and inflation, making it harder for individuals to afford health insurance. However, health insurance costs vary from one individual to another; factors such as age, location and tobacco use are legally allowed to increase the cost of one's premium.

However, individuals with other pre-existing health risks may themselves select costlier healthcare plans that provide greater coverage. Further, other variables may affect tobacco use. Overall, we can categorize factors that potentially drive healthcare cost into two categories: modifiable lifestyle choices and baseline demographic/socioeconomic factors.

Many studies have investigated the role of these variables individually. This study attempts to fill a research gap by investigating how much premium prices can be attributed by baseline demographic/socioeconomic factors (DRF) versus modifiable lifestyle factors (MLC); does one category have a more substantial effect on insurance cost variance? The results indicate how institutions and individuals can make healthcare more affordable. For instance, if insurance cost variance is heavily driven by MLC, customers can lower prices by modifying behaviors. However, if insurance cost is heavily driven by DRF, policy changes may be required to lower costs.

Three variables representing MLC are considered: smoking status (current, former, never), alcohol consumption frequency (never, occasionally, weekly, daily), and BMI. We consider six variables representing DRF, age, sex, highest education level, income, marital status, and household size.

Dataset: We utilize the Medical Insurance Cost Prediction dataset from Kaggle [1]. The dataset contains approximately 100,000 observations including demographic details (Age, Sex, Income) and lifestyle indicators (Smoking status, BMI, Alcohol frequency). Constituent data was derived from three main sources: Centers for Disease Control and Prevention (CDC), U.S. Department of Health & Human Services (HHS), and Centers for Medicare & Medicaid Services (CMS).

Research Question: **How much can premium prices be attributed by baseline demographic/socioeconomic factors (DRF) versus modifiable lifestyle factors (MLC)?**

2. Exploratory Data Analysis

2.1 Summary Statistics

Table 1 summarizes the central tendencies of the continuous variables. The substantial difference between the median (463.58 USD) and mean (582.32 USD) premium indicates a right-skewed distribution, suggesting the presence of high-cost outliers.

Table 1: Summary of Numerical Variables

Average Age	Average BMI	Median Premium	Mean Premium	Max Premium
47.52	26.99	463.58	582.32	10962.55

2.2 Distribution and Transformation

Linear regression assumes approximately normal residuals with constant variance. When the response variable is heavily right-skewed, these assumptions are typically violated. As shown in Figure 1 (Left), the raw annual premium distribution exhibits substantial right-skewness. We therefore applied a log-transformation, $\ln(\text{Premium})$, which produces a more symmetric distribution (Figure 1, Right). This transformation typically improves residual normality and homoscedasticity, which we verify in Section 4.1.

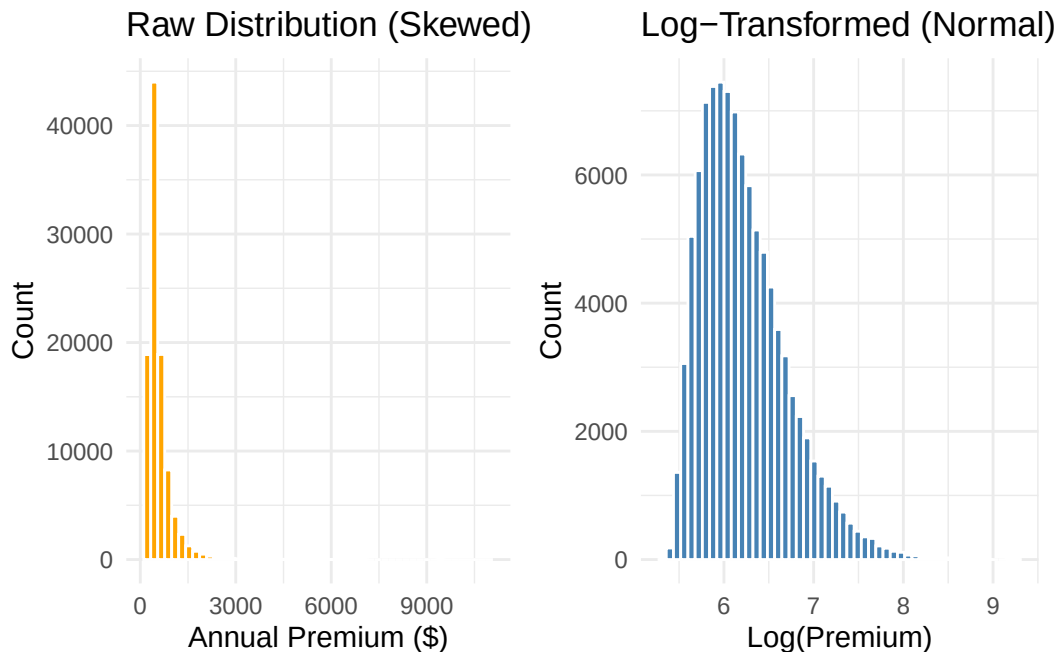


Figure 1: Distribution of annual premiums before and after log-transformation

2.3 Visualizing the “Smoking Penalty” and Interaction

We hypothesized that the cost burden of smoking might vary by age. Figure 2 plots Age against Log-Premium, stratified by smoking status.

The Green Line (Current Smokers) shows a higher intercept, indicating a higher baseline cost. More importantly, the slope appears steeper than that of the Red Line (Never Smokers). This suggests a potential interaction effect: the financial penalty for smoking compounds as the individual ages. We will test this formally in our regression models.

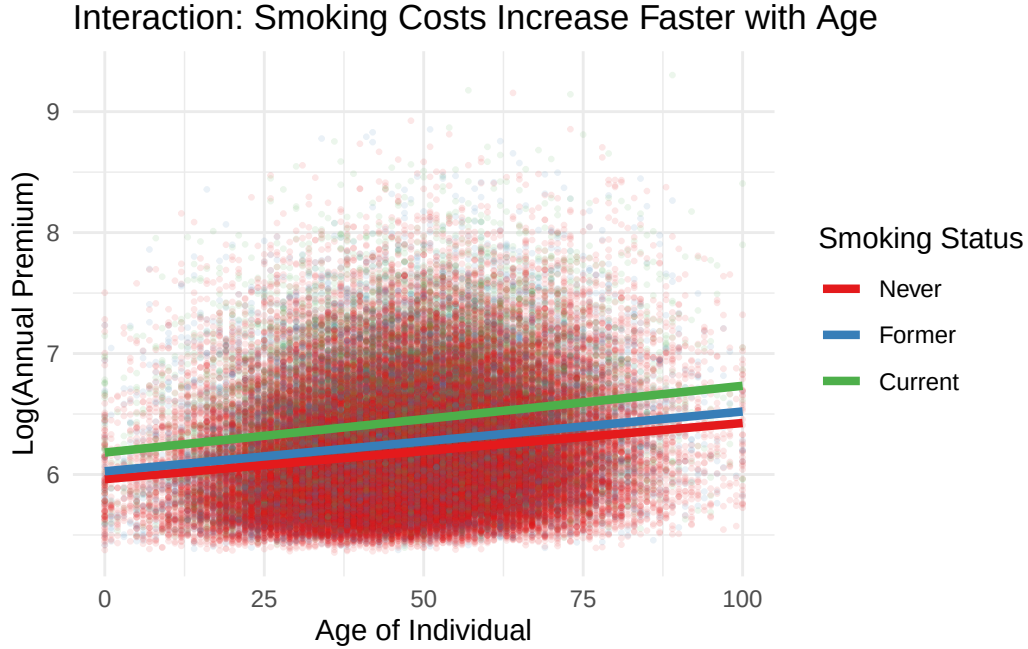


Figure 2: Log-premium by age, stratified by smoking status. The steeper slope for current smokers suggests an interaction effect.

3. Methodology

To isolate the variance explained by lifestyle choices, we employ Hierarchical Multiple Linear Regression[3]. We fit three distinct models:

Model 1: Baseline Demographics This model accounts for fixed factors the individual cannot easily change.

$$\ln(Y) = \beta_0 + \beta_{Age} + \beta_{Sex} + \beta_{Region} + \beta_{Income} + \beta_{MaritalStatus} + \beta_{Urban/Rural} + \beta_{Education} + \epsilon$$

Model 2: Lifestyle This model accounts for only modifiable behaviors that the individual can easily change

$$\ln(Y) = \beta_0 + \beta_{Smoker} + \beta_{Alcohol} + \beta_{BMI} + \epsilon$$

Model 3: Full Model (Demographics + Lifestyle) This model adds modifiable behaviors. Based on the interaction observed in Section 2.3, we include an interaction term for Age and Smoking.

$$\ln(Y) = \beta_0 + \beta_{Age} + \beta_{Sex} + \beta_{Region} + \beta_{Income} + \beta_{MaritalStatus} + \beta_{Urban/Rural} + \beta_{Education} \\ + \beta_{BMI} + \beta_{Smoker} + \beta_{Alcohol} + (\beta_{Age} \times \beta_{Smoker}) + \epsilon$$

We utilize Analysis of Variance (ANOVA) to statistically compare models 1 and 2 with the Full Model. The F-test in the ANOVA will determine if the reduction in the Residual Sum of Squares (RSS) provided by the lifestyle/demographic variables is statistically significant.

4. Results

4.1 Diagnostics and Assumptions Checking

To ensure the validity of our inference, we examined the residual plots for the full Model.

- Linearity: Figure 4 (Left)
- Equal Variance: Figure 4 (Left)
- Independence:
- Normality of Residuals: Figure 4 (Right) shows the Normal Q-Q plot. The points (circles) adhere closely to the diagonal line for the majority of observations. However, the tails deviate upward. This “heavy-tailed” behavior indicates that our model slightly underestimates the premiums for the most extreme, high-cost outliers. Given the large sample size ($N \approx 100,000$), the Central Limit Theorem ensures that our coefficient estimates remain unbiased despite this deviation.

4.2 ANOVA Model Comparison

To test whether the addition of DRF parameters and MLC parameters has an effect on squared residuals of the model, an ANOVA F-test is utilized. The DRF-only model was compared to the full model to assess the effect of MLC parameters; the MLC-only model was compared to the full model to assess the effect

Table 2: ANOVA Comparison: Demographics vs. Full Model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
99980	22900.88	NA	NA	NA	NA
99972	22085.80	8	815.09	461.19	0

Table 3: ANOVA Comparison: Lifestyle vs. Full Model

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
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99993	22683.72	NA	NA	NA	NA
99972	22085.80	21	597.92	128.88	0

Addition of Demographic/Socioeconomic Risk Factors ***H(null)**: $\{Age\} = \{Sex\} = \{Region\} = \{Income\} = \{MaritalStatus\} = \{Urban/Rural\} = _ \{Education\} = 0$ ***H(alt)**: At least one of $\{Age\}$, $\{Sex\}$, $\{Region\}$, $\{Income\}$, $\{MaritalStatus\}$, $\{Urban/Rural\}$, $_ \{Education\}$ $\neq 0$ The test statistic of 461.19 has a F distribution (df1 = 8, df2 = dummy) under the null hypothesis. The corresponding p value (approximately 0) is clearly less than alpha value 0.05; this indicates that we can reject the null hypothesis and conclude that the demographic/socioeconomic parameters are not all equal to zero. Therefore, variation at least one of the socioeconomic/demographic factors seems to help explain the variation in the log(monthly premium); the addition of DRF helps reduce the squared residuals.

Addition of Modifiable lifestyle choices ***H(null)**: $\{Smoker\} = \{Alcohol\} = \{bmi\} = 0$ ***H(alt)**: At least one of $\{Smoker\}$, $\{Alcohol\}$, $\{bmi\}$; $\neq 0$ The test statistic of 139.44 has a F distribution (df1 = 8, df2 = dummy) under the null hypothesis. The corresponding p value (approximately 0) is clearly less than alpha value 0.05; this indicates that we can reject the null hypothesis and conclude that the modifiable life choice parameters are not all equal to zero. Therefore, variation at least one of the modifiable life choices seems to help explain the variation in the log(monthly premium); the addition of MLC helps reduce the squared residuals.

Overall Conclusion: The ANOVA tests conclude that the addition of MLC parameters to a DRF model may help explain the variation in the log(monthly premium). The same result was found when DRF parameters were added to a MLC model. However, as the degrees of freedom are not the same for each ANOVA test, the F value cannot be used directly to compare the impact of the added DRF/MLC variables.

4.3 Comparison of R^2 Values

Table 4: R^2 Values of Demographic, Lifestyle, and Full Model

Category	R2_Value
R^2 Demographic	0.0249303
R^2 Lifestyle	0.0341766
R^2 Full Model	0.0596348

Table 5: ΔR^2 of Demographic and Lifestyle Model Compared to Full Model

Category	R2_Value
ΔR^2 (Full Model - Demographic)	0.0347046
ΔR^2 (Full Model - Lifestyle)	0.0254582

Change in Squared Residuals To compare the effect of adding lifestyle variables versus demographic variables, the R^2 values can instead be observed. R^2 values are extremely low for all three

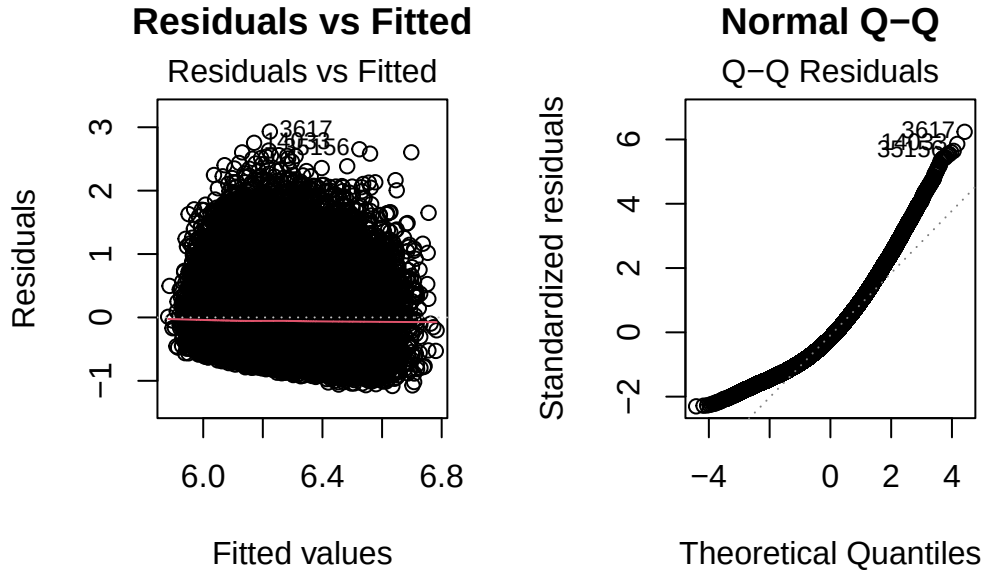


Figure 3

models, at 0.0249, 0.0342, and 0.0596 for the demographic, lifestyle and full model respectively (Table X). This indicates that all three models do not have a strong fit for the data.

The effect of adding lifestyle variables versus demographic variables can be assessed by comparing the change in R^2 value of the lifestyle and demographic model with the full model. The demographic-only model has a R^2 that is 0.0347 less than the full model (addition of lifestyle variables), while the lifestyle model has an R^2 that is 0.0255 less than the full model (addition of demographic variables). This indicates that the addition of the lifestyle variables has a more substantial effect on reducing the squared residuals of the model.

4.4 Regression Coefficients

Table 3 presents the coefficients for the Full Model. Since the response is log-transformed, coefficients (β) can be interpreted as an approximate $100 \times \beta\%$ change in the premium for small values.

Table 6: Coefficients for Modifiable Lifestyle Factors (Model 2)

Predictor	Estimate	Std. Error	P-Value
BMI	0.00562	0.0002976	<0.001
Smoker: Former	0.06373	0.0123031	<0.001
Smoker: Current	0.22166	0.0143493	<0.001
Alcohol: Occasional	-0.00062	0.0034994	0.859
Alcohol: Weekly	-0.00388	0.0042996	0.367
Alcohol: Daily	0.01245	0.0071756	0.083

age:Smoker: Former	0.00032	0.0002462	0.193
age:Smoker: Current	0.00087	0.0002870	0.002

Key Findings:

1. Smoking: Being a “Current Smoker” has a large positive coefficient (0.221), indicating a substantial premium increase compared to never-smokers, holding other factors constant.
2. BMI: The coefficient for BMI is positive and statistically significant ($p < 0.001$). Higher body mass index is reliably associated with higher insurance costs.
3. Alcohol: The p-values for alcohol frequency are > 0.05 . This suggests that, unlike smoking and BMI, self-reported alcohol frequency is not a statistically significant predictor of insurance premiums in this dataset.
4. Interaction (Age $\times$ Smoker): The significant interaction term ($p = 0.002$) confirms our EDA finding. The slope of the cost curve is steeper for smokers. This means the dollar-cost of smoking increases as the individual ages.

5. Discussion

6. References

[1] Kaggle. (2024). Medical Insurance Cost Prediction Dataset. [2] Wooldridge, J. M. (2015). Introductory Econometrics. [3] UVa Library. (2023). Hierarchical Linear Regression. <https://library.virginia.edu/data/articles/hierarchical-linear-regression> [4] American Medical Association. (2024). National Health Expenditure Data. <https://www.ama-assn.org/> [5] Bureau of Labor Statistics. (2023). Consumer Expenditure Survey: Health Insurance. [6] HealthCare.gov. (2024). How Insurance Companies Set Health Premiums. <https://www.healthcare.gov/how-plans-set-your-premiums/> [7] Kaiser Family Foundation. (2023). Employer Health Benefits Survey. <https://www.kff.org/>